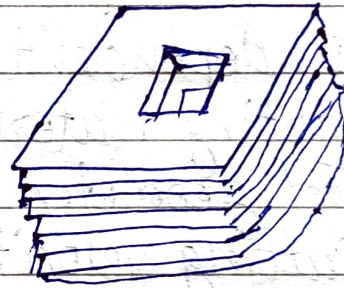
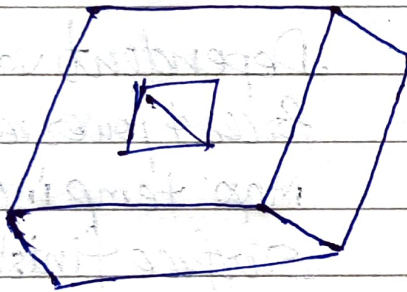


- A. Shaping and forming (zero process), no change in mass.
- B. Joining process (positive process) $m \uparrow$
- C. Removal process (Negative process) $m \downarrow$
- D. Regenerative Manufacturing (Rapid Prototyping).



1. Production of solid products in layer by layer from raw materials in different forms.
2. It is very Rapid, accurate and used for rapid prototyping at tooling.
- 3.



Adv.

- Process is independent of part feature.
- No blanks are required.
- Toolless process
- Easily automation possible.

Machining:

It is an essential process of finishing by which jobs are produced to the desired dimensions and surface finish by gradually removing the excess material from the preformed blank in the form of chip with the help of

cutting tool moved past the work surface.

- Machining is a removal process.

Machining aim to

- Fulfill its functional requirement
- Improve its performance
- prolong its service.

Drawback

- Loss of material in the form of chip.

IAS 2009 main

Q. Name of four independent & three dependent variable in metal cutting.

Independent.

Starting material (work/tool)

Tool geometry:

Cutting velocity

Lubrication.

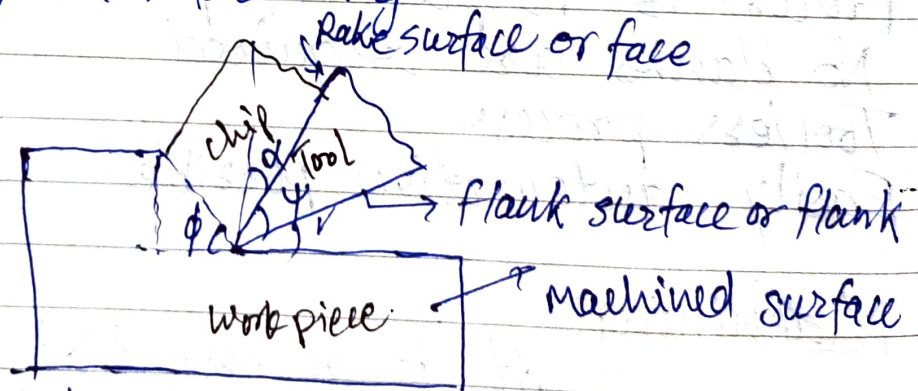
Dependent variable

forces & power required

Max temp in cutting

Surface finish

Orthogonal Machining.



Rake angle $\rightarrow \alpha$

Flank angle $\rightarrow \phi$

$\psi \rightarrow$ Lip angle or wedge angle or cutting angle or knife angle

$\phi \rightarrow$ shear angle

Thickness of chip

$t_c \rightarrow$ chip thickness

$t \rightarrow$ uncut or undeformed chip thickness

$$r = \frac{t}{t_c} \quad r < 1 \quad t < t_c$$

t_c is increased due to shear failure
 $t_c > t$ becoz of shear failure.

- There is no difference b/w depth of cut & uncut chip thickness in orthogonal machining only. $\Rightarrow$ $t = d$
- For turning $t = d / \sin \gamma$

Rake Surface: - The surface along which chip moves upward, is called rake surface of tool.

Flank or flank surface:

The other surface which is relieved to avoid rubbing with the machined surface is called flank or flank surface.

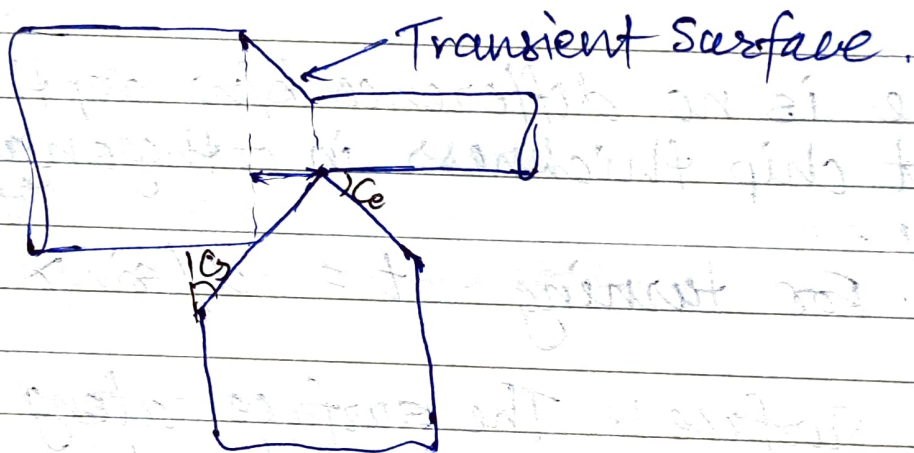
Rake angle -

Angle of inclination of rake surface from reference plane i.e normal to horizontal machined surface.

Clearance Angle or relief angle (γ):
Angle of inclination of clearance or flank surface from the finished (machined surface)

Turning:-

Cutting Condⁿ for turning.



$$\begin{aligned} \text{Speed } V &= \pi D N \text{ mm/min} \\ &= \frac{\pi D N}{60} \text{ mm/s.} \\ &= \frac{\pi D N}{1000} = \text{m/min.} \end{aligned}$$

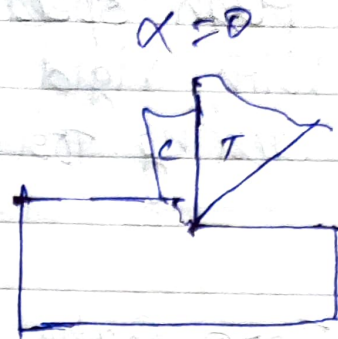
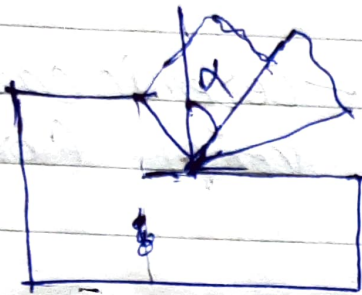
Feed is used for position the tool.
 $f = \text{mm/rev.}$

$$d = \text{depth of cut} = \frac{D_1 - D_2}{2}$$

$$\boxed{D_2 + 2d = D_1}$$

Discussion on Rake angle.

- Rake angle can be +ve, -ve or zero



Positive Rake.

→ [sharpness more]

- Reduce cutting force
- Reduce cutting power
- ~~sh~~ Positive Rake angle are recommended for-
- Machining low strength material.
- Low power machining
- Long shaft of small diameter
- Set up lacks strength and rigidity.
- Low cutting speed.

VIMP Cutting tool material → High speed steel
18% Tungsten, 4% Cr, 1% V

Negative Rake.

- Increase edge strength (Mechanical & thermal)
- Increase life of the tool. Tool life ↑
- Increase the cutting force
- Requires high cutting speed.
- Requires Ample (very high) power.
- Heavy impact loads.

Negative Rake angle are recommended -

- Machining high strength alloy.
- High speed cutting
- With rigid set-up
- Cutting tool material - Ceramic, carbide (comp), (brittle)

Zero Rake.

- To simplify design and manufacturing of the form of tool.
- Increase tool strength $\uparrow$ Tool Strength
- Avoids digging of tool into workpiece.
- Brass is turned with zero rake angle
- CI used zero rake angle.

Brass & CI $\rightarrow \alpha = 0$

Clearance angle: — $3-15^\circ$

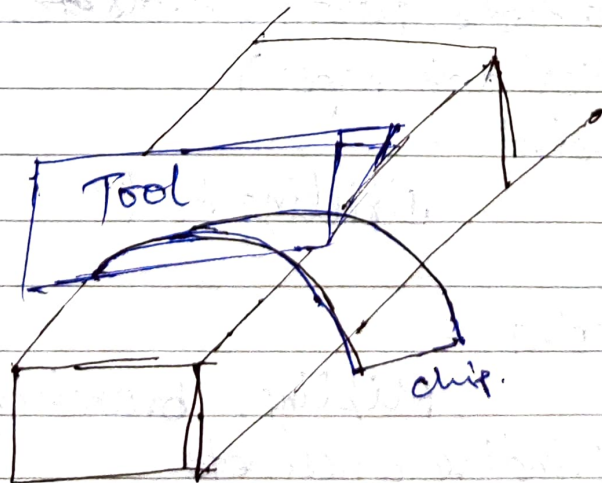
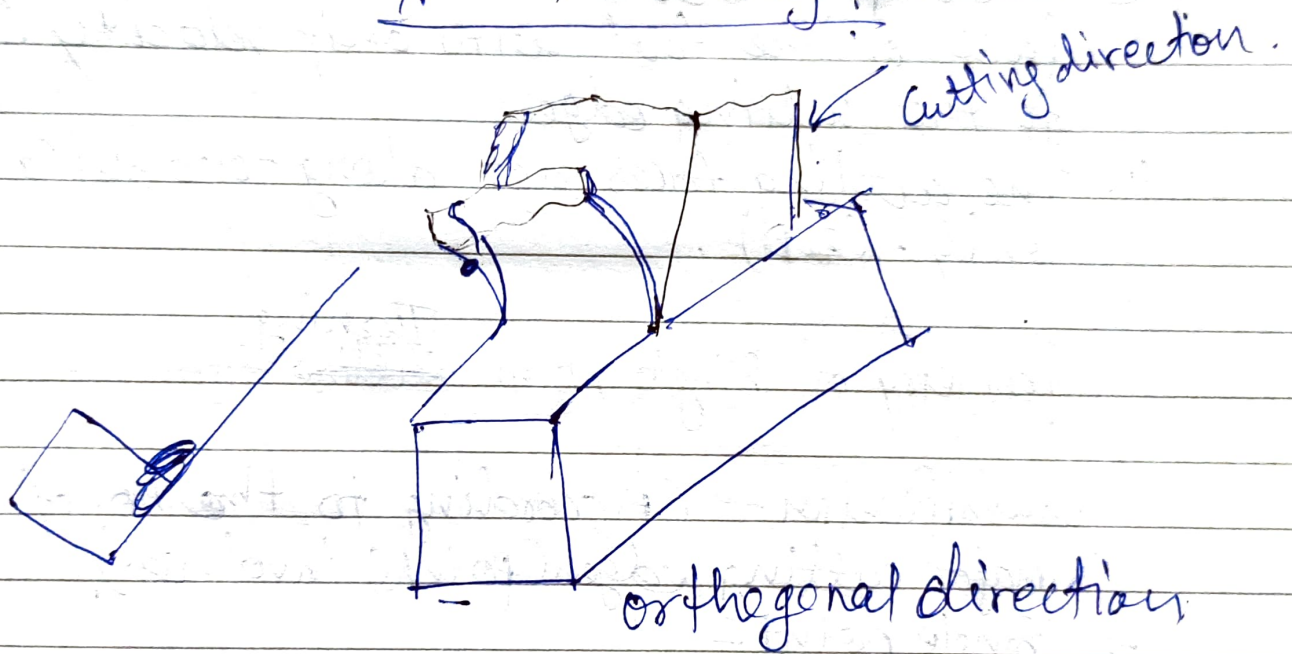
→ provided to avoid rubbing of the tool (flank) with the machined surface.

→ Reduce tool wear

→ Increase tool life. Tool life $\uparrow$, wear $\downarrow$

→ Must be positive ($3^\circ-15^\circ$)

Types of Machining:



Oblique Cutting

Orthogonal Cutting

1. Cutting edge of the tool is perpendicular to the dirⁿ of cutting velocity.
2. The cutting edge is wider than the workpiece width and extends beyond the workpiece on either side. Also the width of the workpiece is much greater than the depth of cut.
3. The chip generated flows on the rake face of the tool with chip velocity $\perp$ to the cutting edge.
4. The cutting forces act along two dirⁿs only. (2D).

Geometry of Single point ^{Turning} ~~Cutting~~ tool: -

Classification - (According to the no. of major cutting edges (points) involved.)

→ Single point -

Turning, shaping, planning, slotting tools, Parting tools etc.
(vertical axis shaper)

→ Double point -

Drilling tool

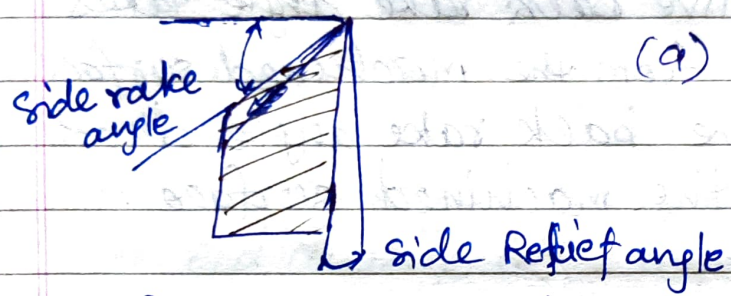
→ Multipoint -

Milling, broaching, hobbing tools
saw, grinding wheel etc.

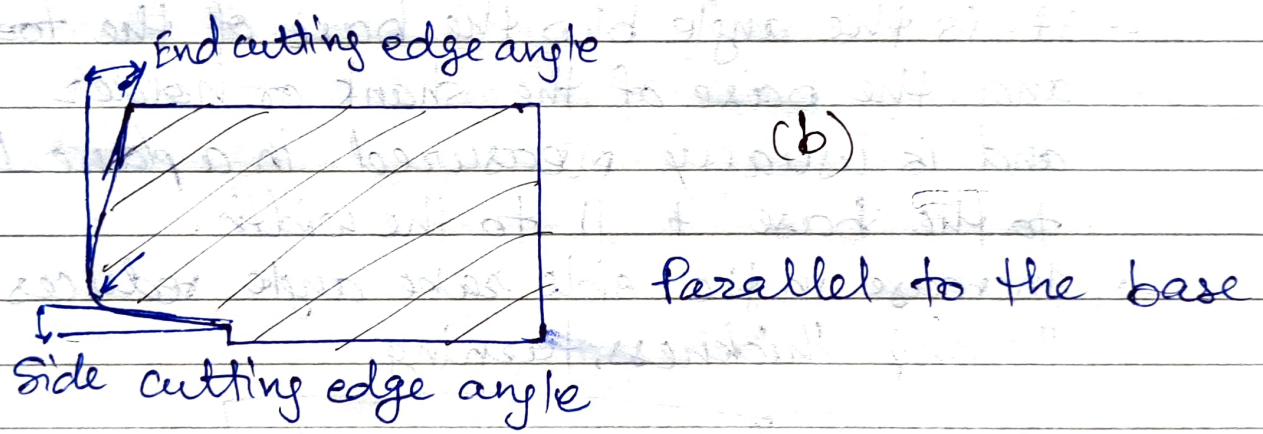
System of Description of Tool Geometry.

- 1. M/c reference system - ASA or ANSI
- 2. Tool reference system - ORS ^{ISO (old)} and NRS ^{ISO (new)}
- 3. Work reference system - WRS

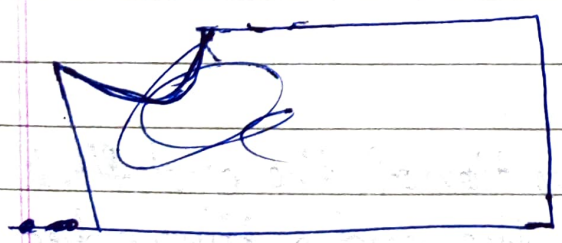
ASA (Single point cutting tool), for description



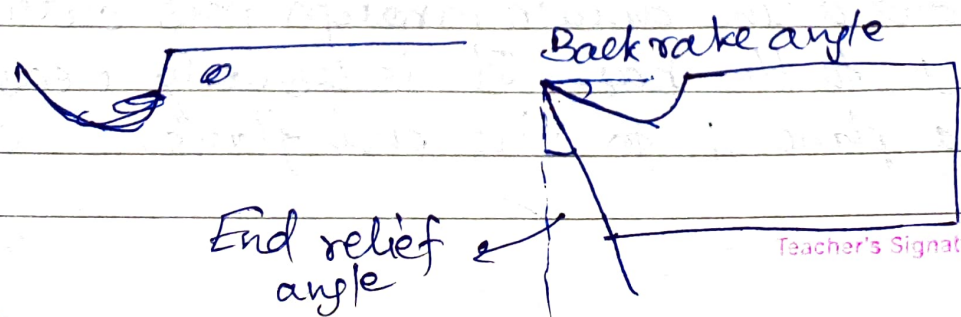
(Parallel to the width & perpendicular the base)



Parallel to the base



perpendicular the base & parallel to the length



Back rake angle (α_b) -

- It is the angle b/w the face of the tool and the base of the shank or holder & is usually measured in a plane $\perp$ to the base & $\parallel$ to the length of the tool.
- It affects the ability of the tool to shear the work material and form the chip.
- In turning positive back rake angle takes the chips away from the machined surface whereas negative back rake angle direct the chip on to the machined surface.

Side rake angle (Axial rake) (α_s) -

- It is the angle b/w the base of the tool and the base of the shank or holder, and is usually measured in a plane $\perp$ to the base & $\parallel$ to the width.
- Increase in the side rake angle reduces the chip thickness, turning.

$$\boxed{\alpha_s \uparrow \Rightarrow \frac{dt}{d\alpha_s} \quad \uparrow r = \frac{t}{t_0}}$$

End relief angle (γ_e)

- It is the angle b/w the portion of the end flank immediately below the end cutting edge and a line drawn through this cutting edge $\perp$ to the base. It is usually measured in a plane $\perp$ to the end flank.

→ End relief angle prevents friction on the flank of the tool.

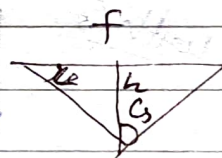
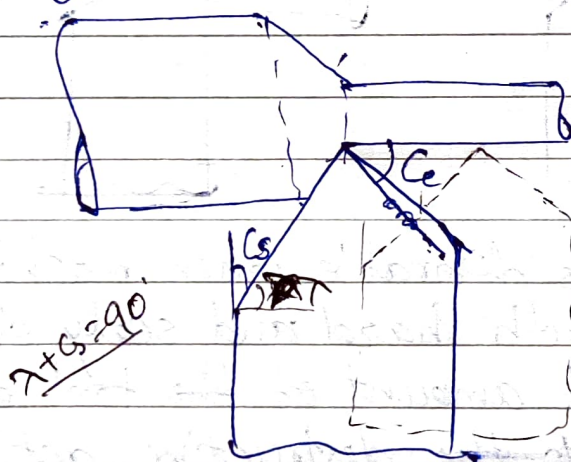
Side relief angle (λ_s)

→ It is the angle b/w the portion of the side flank immediately below the side cutting edge, and a line drawn through this cutting edge $\perp$ to the base.

→ It is measured in a plane $\perp$ to the side flank.

End cutting edge angle, ECEA (C_e) —

→ The end cutting edge angle is the amount that the end-cutting edge slopes away from the nose of the tool, so that it will clear the finished surface on the work piece, when cutting with the side cutting edge.



$C_e \uparrow \Rightarrow$ Weak Tool

$8^\circ - 15^\circ$

→ It prevents the trailing end of the cutting edge of the tool from rubbing against the work-piece.

→ A larger end cutting edge angle weakens the tool.

→ It's usually kept b/w 8° to 15°

Side cutting edge angle, SCEA (ϕ_s)

- It is the angle which prevents interference as the tool enters the work material. (Normally $15-30^\circ$).

→ larger this angle, the greater the component of force tending to separate the work and the tool. (may induce chatter).

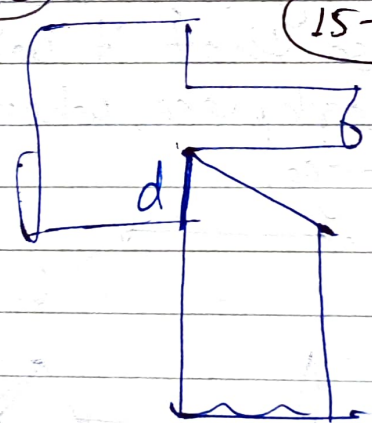
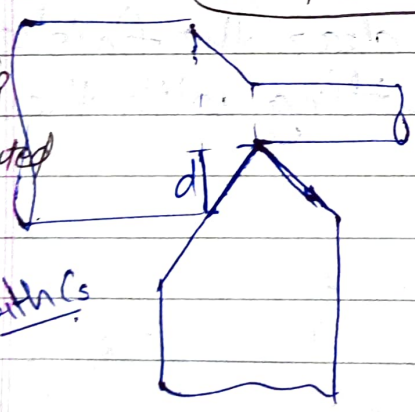
→ At its increased value it will have more of its length in action for a given depth of cut.

→ At its increased value it produce thinner and wider chip that will distribute the cutting heat. (increase tool life)

$\phi_s \uparrow \Rightarrow \text{force} \uparrow$
 $\phi_s \uparrow \Rightarrow \text{Thin + wide chip}$
 Heat Distributed
 Tool life $\uparrow$
with ϕ_s

(No influence on F_c)

($15-30^\circ$)



Zero ϕ_s
 Casting & forging

→ Zero SCEA is desirable when machining casting & forging with hard and scaly skins, because at the least amount of the tool edge should be exposed to the destructive action of the skin.

→ SCEA has no influence on cutting force or cutting power consumption.

Angle in drill generally 118°

F_y or Radial force will increase with SCEA

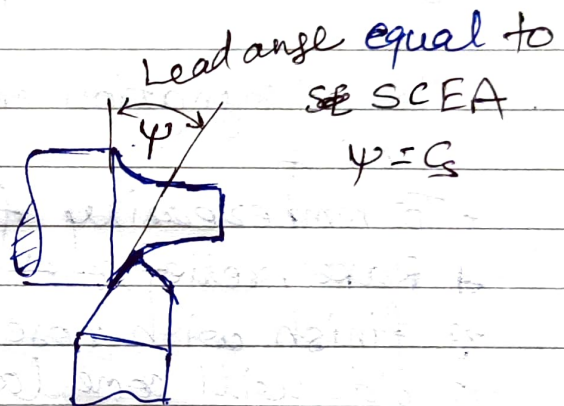
$$F_y \uparrow = f_f \cos \lambda = f_f \sin \phi \uparrow$$

F_x or Feed force will decrease with SCEA

$$F_x \downarrow = f_f \sin \lambda = f_f \cos \phi \uparrow \quad (\lambda + \phi = 90^\circ)$$

$\lambda \rightarrow$ principle cutting edge angle.

SCEA and Lead angle



depth of cut

Lip angle -

$\psi \uparrow \Rightarrow$ hard metals, $d \uparrow$, Tool life $\uparrow$ or wedge angle

- $\rightarrow$ Lip angle or cutting angle or knife angle depends on the rake and clearance angle provided on the tool and determine the strength of the cutting edge.
- $\rightarrow$ A larger lip angle permits machining of harder metals, allow heavier depth of cut, increase tool life, better heat dissipation.

Nose Radius $\rightarrow$

- $\rightarrow$ It is curvature of the tool tip.
- $\rightarrow$ It strengthen the tool nose by reducing stress concentration.

$K_t \downarrow$

→ It provide better surface finish.

$$\boxed{h_f = \frac{f^2}{8RT}} = \frac{f}{\tan \phi + \cot \phi}$$

- But too large a nose radius will induce chatter.
- If nose radius increased cutting force and cutting power increased.
- It increases the life of the tool.

Tool designation or Tool signature (ANSI)
or ASA

- To remember easily follow the rule.
- 1. Rake, relief, cutting edge
- 2. finish with nose radius (inch)
- 3. Side will come last.

$$\boxed{R - \alpha - r - r_1 - C_e - C_s - R}$$

Orthogonal Rake System (ORS), ~~for~~

for calculation

$$\boxed{i - \alpha - r - r_1 - C_e - \lambda - R}$$

↓ Greater than 60°

- Inclination angle (i)
- Orthogonal rake angle (α)
- Side relief angle (r)
- End relief angle (r_1)
- End cutting edge angle (C_e)
- Principal cutting edge angle or Approach angle ($\lambda = 90 - C_s$)
- Nose radius (R) (mm)

- For orthogonal cutting, $i = 0$
- For oblique cutting, $i \neq 0$

Inter-conversion b/w ASA & ORS. (IES Generally).

$$\tan \alpha = \tan \alpha_s \sin \lambda + \tan \alpha_n \cos \lambda \quad \text{Imp.}$$

$$\tan \alpha_n = \cos \lambda \tan \alpha + \sin \lambda \tan i$$

$$\tan \alpha_s = \sin \lambda \tan \alpha - \cos \lambda \tan i$$

$$\tan i = -\tan \alpha_s \cos \lambda + \tan \alpha_n \sin \lambda$$

Critical correlations:

$$\text{When } \lambda = 90^\circ \quad \alpha_n = \alpha \quad \Rightarrow \text{Imp.}$$

$$\text{When } i = 0 \quad \alpha_n = \alpha$$

$$\text{When } i = 0 \text{ \& } \lambda = 90^\circ \quad \alpha_n = \alpha_s = \alpha \quad \Rightarrow \alpha_s = 0$$

(Pure orthogonal cutting).

λ is principal cutting edge angle @ Approach Angle

i is inclination angle

α_n normal rake angle (NRS)

α_s is side rake angle (ASA)

α is orthogonal rake angle (ORS)

Shear angle (ϕ) :- cutting parameter

$$\frac{r}{r_c} = \frac{t}{t_c} = \frac{L}{L_c} = \frac{V_c}{V} = \frac{\sin \phi}{\cos(\phi - \alpha)} = \frac{1}{h}$$

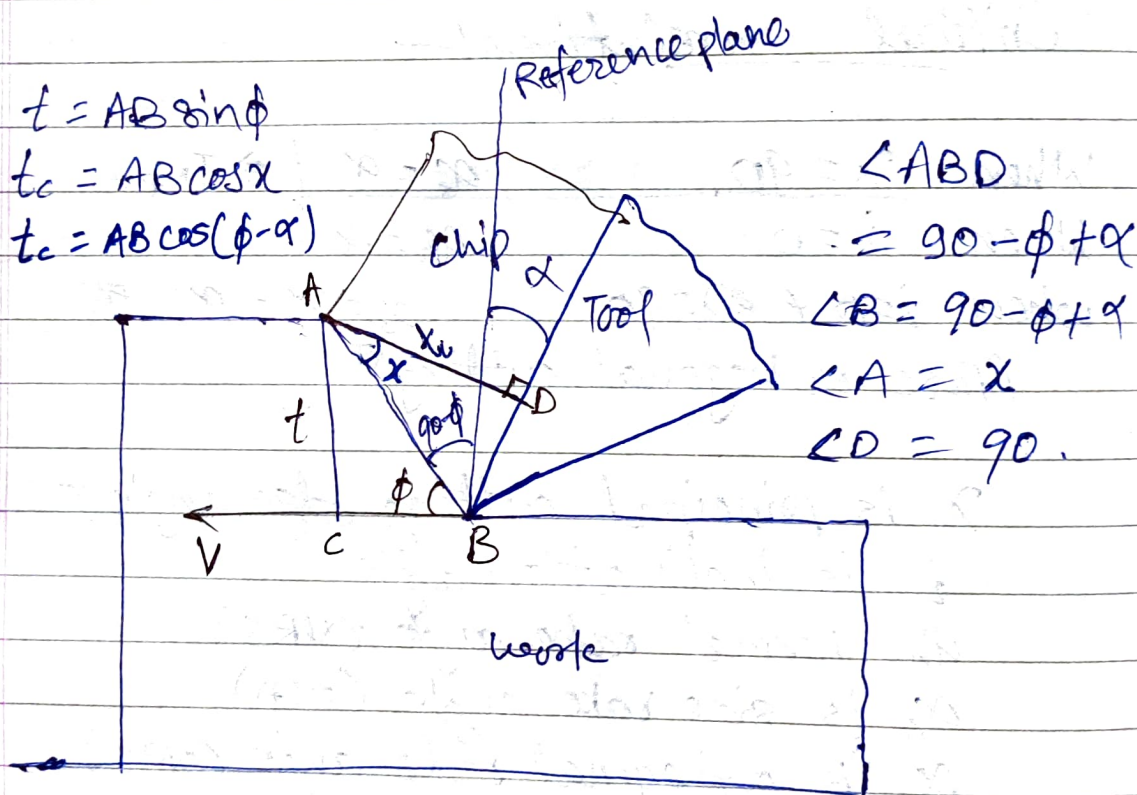
$$\text{and } \boxed{\tan \phi = \frac{2 \cos \alpha}{1 - 2 \sin \alpha}} \quad \text{Imp.}$$

where

r = Chip thickness ratio or cutting ratio ; $r < 1$
 $h = 1/r$ = Inverse of chip ratio or chip reduction factor or chip compression ratio
 $h > 1$.

• If $\alpha = 0$; $\tan \phi = r$

Prove : $\tan \phi = \frac{r \cos \alpha}{1 - r \sin \alpha}$ [IAS main 2009]



$$180 = 90 - \phi + \alpha + x + 90$$

$$x = \phi - \alpha$$

$$r = \frac{t}{t_c} = \frac{AB \sin \phi}{AB \cos(\phi - \alpha)}$$

$$r = \frac{\sin \phi}{\cos(\phi - \alpha)}$$

$$\sin \phi = r \cos(\phi - \alpha)$$

$$\text{or } \sin \phi = r \cos \phi \cos \alpha + r \sin \phi \sin \alpha$$

$$\text{or } \sin \phi - r \sin \phi \sin \alpha = r \cos \phi \cos \alpha$$

$$\text{or } \sin \phi (1 - r \sin \alpha) = r \cos \phi \cos \alpha$$

$$\text{or } \frac{\sin \phi}{\cos \phi} = \frac{r \cos \alpha}{1 - r \sin \alpha}$$

$$\text{or } \boxed{\tan \phi = \frac{r \cos \alpha}{1 - r \sin \alpha}}$$

4. A single-point cutting tool with 12° rake angle is used to m/c a steel work piece. Depth of cut i.e. uncut thickness is 0.81 mm . The ch.

$$t_c = 1.8 \text{ mm}, \quad \phi = ?$$

$$\begin{aligned} t &= 0.81 \\ t_c &= 1.8 \end{aligned} \quad r = \frac{t}{t_c} = \frac{0.81}{1.8} = 0.45 = \frac{\sin \phi}{\cos(\phi - \alpha)}$$

$$\tan \phi = \frac{0.45 \cos 12}{1 - 0.45 \sin 12} \quad \phi = \underline{26^\circ}$$

Cutting shear strain (ϵ)

$$\boxed{\begin{aligned} \epsilon &= \frac{\cot \phi + \tan(\phi - \alpha)}{\cos \alpha} \\ &= \frac{1}{\sin \phi \cos(\phi - \alpha)} \end{aligned}}$$

• If $\alpha = 0$, then

$$\epsilon = \cot \phi + \tan \phi \geq 2 \quad \left| x + \frac{1}{x} \text{ form} \right.$$

$$AM \geq GM$$

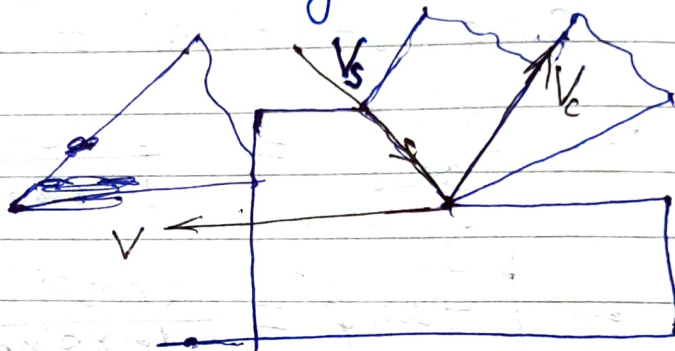
$$\frac{x + \frac{1}{x}}{2} \geq \sqrt{x \cdot \frac{1}{x}}$$

$$\text{or } x + \frac{1}{x} \geq 2$$

Teacher's Signature

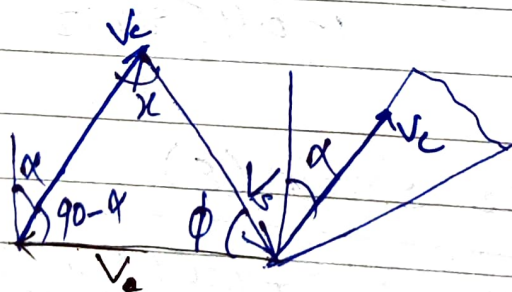
$E_{min} = 2$ when $\phi = 45^\circ$

Velocities in metal cutting-



- The velocity of the tool relative to the work-piece (V) is called the cutting speed.
- The velocity of the chip relative to the work piece (V_s) is called the shear velocity.
- The velocity of the chip relative to the tool (V_c) is called the chip velocity.

Derive the expression for velocities in metal cutting.



$$\alpha + \phi + 90 - \alpha = 180$$

$$\text{or } \alpha = 90 - \phi + \alpha$$

$$\alpha = 90 - (\phi - \alpha)$$

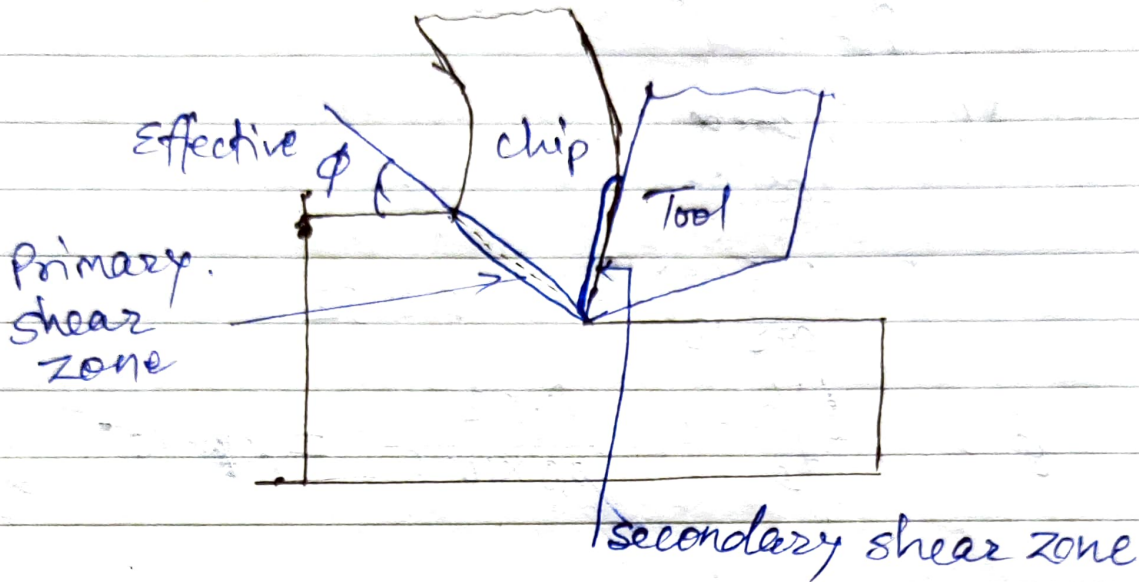
Sine Rule

$$\frac{V}{\sin(90 - (\phi - \alpha))} = \frac{V_c}{\sin \phi} = \frac{V_s}{\sin(90 - \alpha)}$$

$$\boxed{\frac{V}{\cos(\phi - \alpha)} = \frac{V_c}{\sin \phi} = \frac{V_s}{\cos \alpha}}$$

$$\frac{V_e}{V} = \frac{\sin \phi}{\cos(\phi - \alpha)} = r$$

$$\frac{V_s}{V} = \frac{\cos \alpha}{\cos(\phi - \alpha)}$$



Shear strain Rate.

(Note: It is not shear strain it is rate of shear strain i.e. flow)

$$\dot{\epsilon} = \frac{d\epsilon}{dt} = \frac{V_s}{\text{thickness of shear zone } (t_s)}$$

- Thickness of shear zone can be taken as $1/10^{th}$ (10% of shear plane length and its max^m value is 25 microm)
- Unit 1/sec.

2. $r = 0.4$, $t = 0.6 \text{ mm}$, $\alpha = 10^\circ$, $V = 2.5 \text{ m/s}$, $t_s = 25 \mu$

~~$0.4 = \frac{V_{\cancel{0.6}}}{\cancel{0.6}}$~~ $\Rightarrow t_c = \frac{0.6}{0.4} = 1.5$ $\dot{\epsilon} = ?$

~~$V_c = 0.24$~~

$$\tan \phi = \frac{r \cos \alpha}{1 - r \sin \alpha}$$

$$= \frac{0.4 \cos 10}{1 - 0.4 \sin 10} = 0.42$$

$$\phi = 22.9^\circ \approx 23^\circ$$

$$V_s = \frac{V \cos \alpha}{\cos(\phi - \alpha)}$$

$$= \frac{2.5 \cos 10}{\cos 13^\circ} = 2.52$$

$$\dot{\epsilon} = \frac{V_s}{t_s} = \frac{2.52}{25 \times 10^{-6}} = 1.01 \times 10^5 \text{ s}^{-1}$$

Cause of chip formation -

- Shear Yielding - in ductile material
- Brittle fracture - in brittle material

Types of chip.

- Continuous chip
- Discontinuous chip
- Continuous chip with BUE
- Serrated chip

→ Conditions for forming Discontinuous chip.
of irregular size and shape
work material brittle (such as gray cast iron)

of regular size and shape
work material - ductile but hard and work
hardenable

| High carbon steel like mild steel.

feed - large
 tool rake - negative
 cutting fluid - absent or inadequate
 Using multipoint cutting tool like milling, broaching.

→ Conditions for forming Continuous chip without BUE

- Work material - ductile
- Cutting velocity - high
- Feed rate - Low
- Rake angle - positive and high
- Cutting fluid - both cooling and lubrication.

→ Conditions for forming Continuous chip with BUE

- Work material - ductile
- Cutting velocity - Medium
- Feed - Medium
- Cutting fluid - absent or inadequate.

→ Built-Up-Edge (BUE) formation -

- In machining ductile material with long chip tool contact length, lot of stress and temperature develops in the secondary deformation zone at the chip tool interface.
- Under such high stress & temp in b/w two clean surfaces of metals, strong bonding may locally take place due to adhesion similar to welding.
- Such bonding will be encouraged and accelerated if the chip tool materials have mutual affinity or solubility.

- The weld material starts forming as an embryo at the most favourable location and thus gradually grows.
- With the growth of the BUE, the force also gradually increase due to wedging action of the tool tip along with the BUE formed on it.
- Whenever the force exceeds the bonding force of the BUE, the BUE is broken or sheared off and taken away by the following chip. The again starts forming and grow.
- Low cutting speed also contributes to the formation of BUE.
(Milling, Broaching are prone to BUE formation due to low speed).

Effects of BUE formation:

Harmful effect

- Poor surface finish.
- It unfavourably changes the rake angle at the tool tip causing increase of cutting force i.e. power consumption.
- Induce vibration. (Dimensions & accuracy affected)

Good effect

- BUE protects the cutting edge of the tool i.e. increase tool life.

Reduction or Elimination of BUE

Increase

Cutting speed ↑

Rake angle ↑

Decrease

Feed $\downarrow$

Depth of cut $\downarrow$

Use

- Cutting fluid
- Change cutting tool material (as cermets).

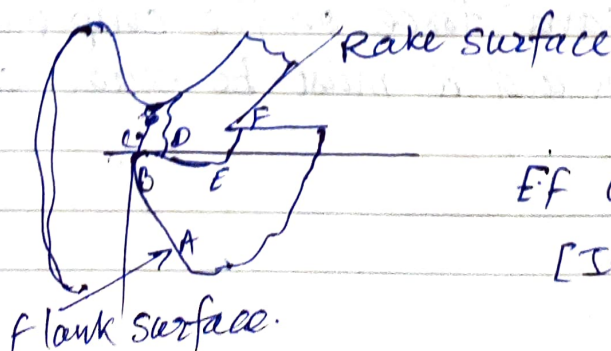
→ Serrated Chips. (Semi continuous). → Ti $K\downarrow$

- Serrated chips also called segmented or non-homogeneous chips are semi-continuous chips with zones of low and high shear strain.
- Metals with low thermal conductivity & strain strength that decreases sharply with temp., such as titanium exhibit this behaviour, the chips have sawtooth like appearance.

When is forced chip breaking necessary and why?
When chips continuously form and come out very hot, sharp and at quite high speed.

Under the condition

- Soft ductile work material
- Flat rake surface with positive or near zero rake.
- For
- Safety and convenience of the operator.
- Easy collection and disposal of chips.



EF chip breaker

[Integrated chip breaker]

Teacher's Signature